AerE 421

Lab 3 Report

10/08/24

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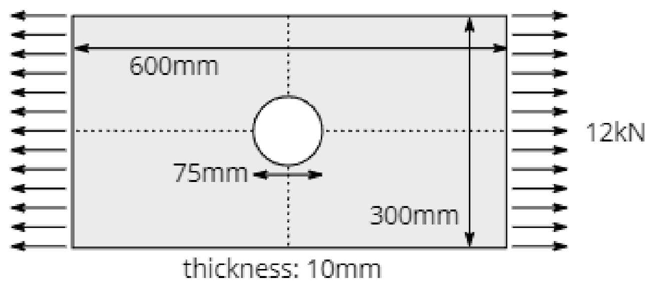
Part 1 Problem Statement

1 Plate with single hole under uniaxial tension

Consider a plate made of 6061 Aluminum subjected to a plane-strain tensile load as shown in the following figure.



Determine the stress concentration factor induced by the hole in the plate, and compare to the prediction from stress concentration factor formulae. To complete this part of the lab, you must report the following:

1. **Computational model**: describe the geometry and mesh of the final model.
   (a) A figure showing the mesh with loads and constraints.
   (b) The final element type you used and a description of it. (Recommended: Plane182, Plane183.)
2. **Results**: report results from the final solution.
   (a) Contour plots of $\sigma_{xx}, \sigma_{yy}, \sigma_{xy}$
   (b) Contour plot of von Mises stress
3. **Analysis**: report the final stress concentration factor and compare to the prediction from the stress concentration factor formula (you need to look this up).
4. **Error estimation**: you must demonstrate that your solution is not dependent on neither the size of mesh nor the order of the element. For this, you must show that the maximum stress converges to a constant value under increased mesh resolution (h-refinement) and increased element order (p-refinement).

   Plot the relative error (based on the theoretical value) in the maximum predicted stress vs # of nodes for each element type. Use logarithmic scaling on your axes, and determine whether the final reported solution is sufficiently accurate.

1. Computational Model

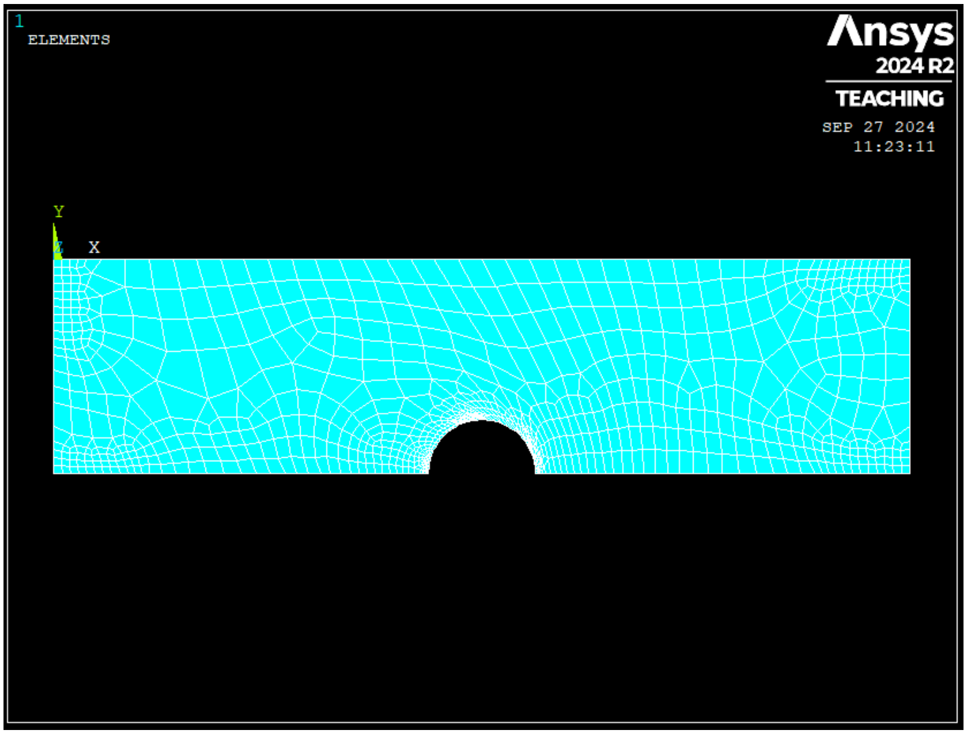


Figure 1.1. Plate modeled with mesh grid

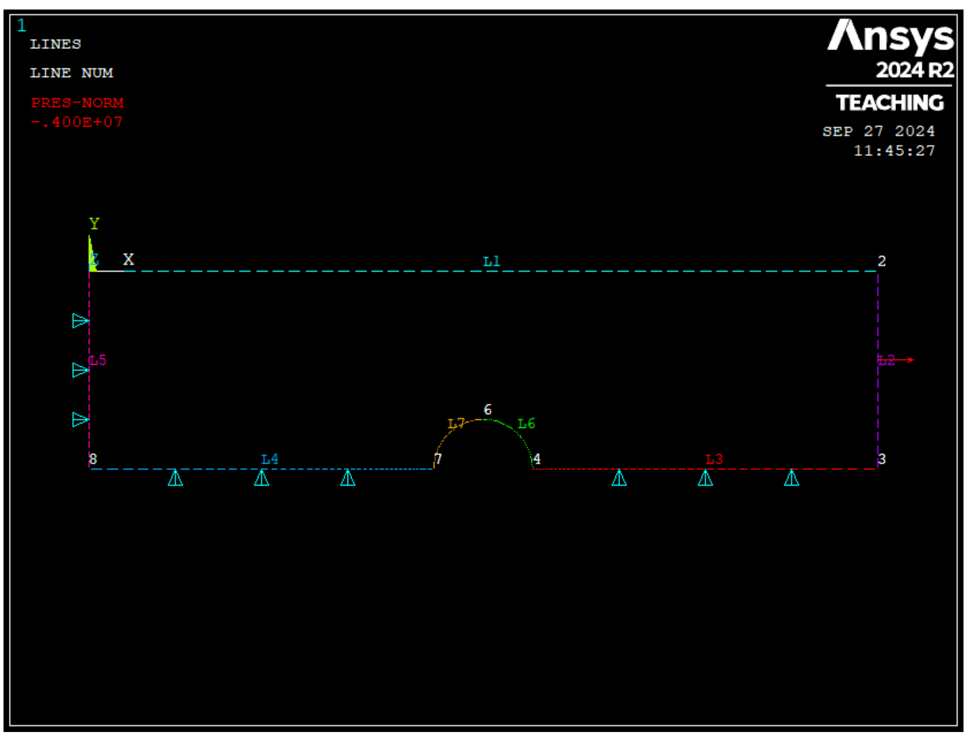


Figure 1.2. Plate constraints and applied loads

For our plate model, we used element type Plane182. It can be used as a plane element (plane stress, plane strain, or generalized plane strain). This element type has plasticity, hyperelasticity, stress stiffening, large deflection, and large strain capabilities. It has a mixed-formulation capability for simulating deformations of nearly incompressible elastoplastic materials, and fully incompressible hyperelastic materials.

2. Results

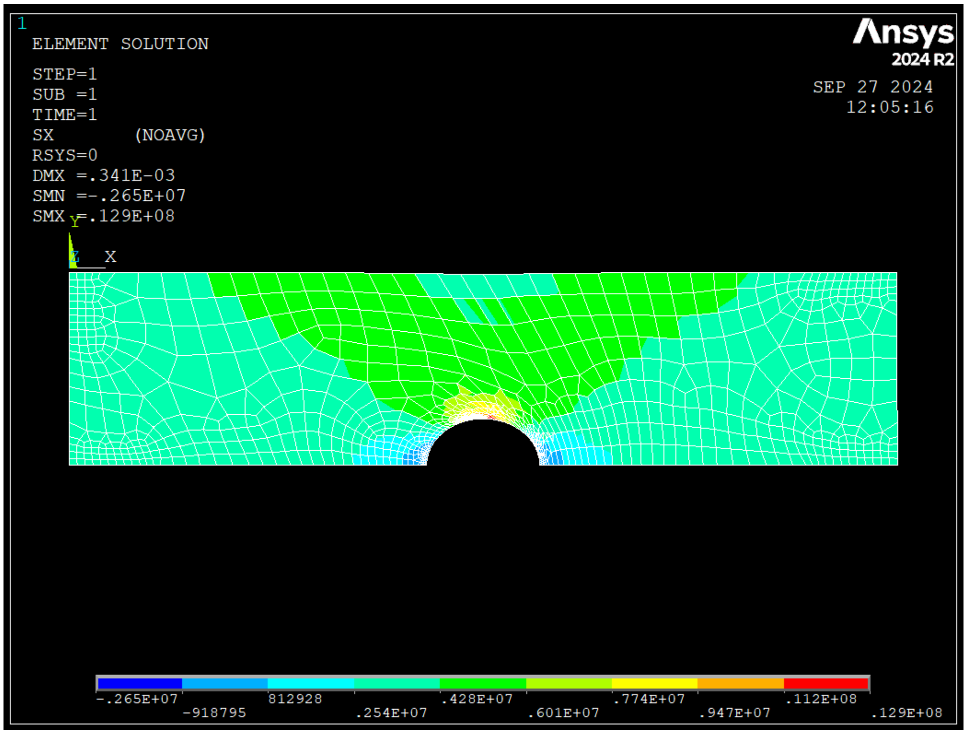


Figure 1.3. Axial stress in X-direction (Pa)

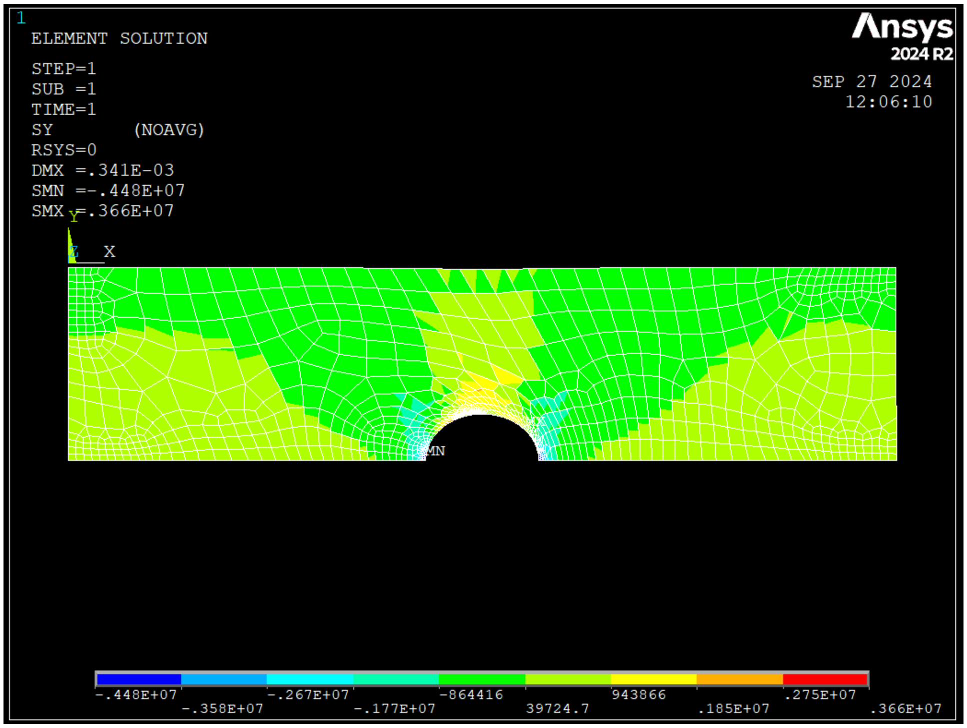


Figure 1.4. Axial stress in Y-direction (Pa)

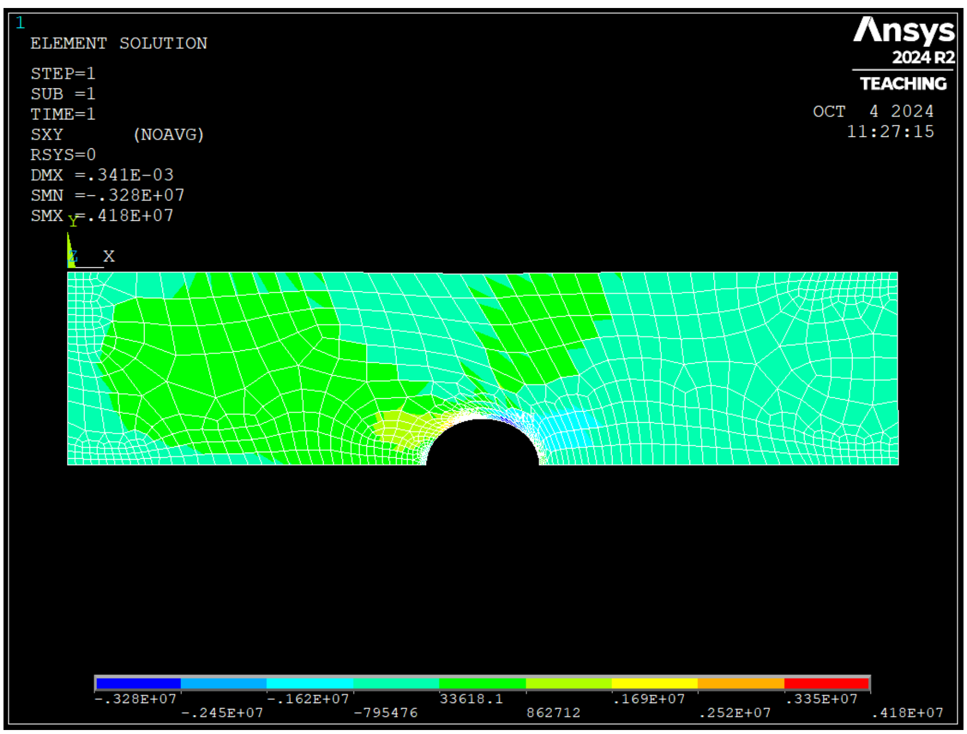


Figure 1.5. Shear stress (Pa)

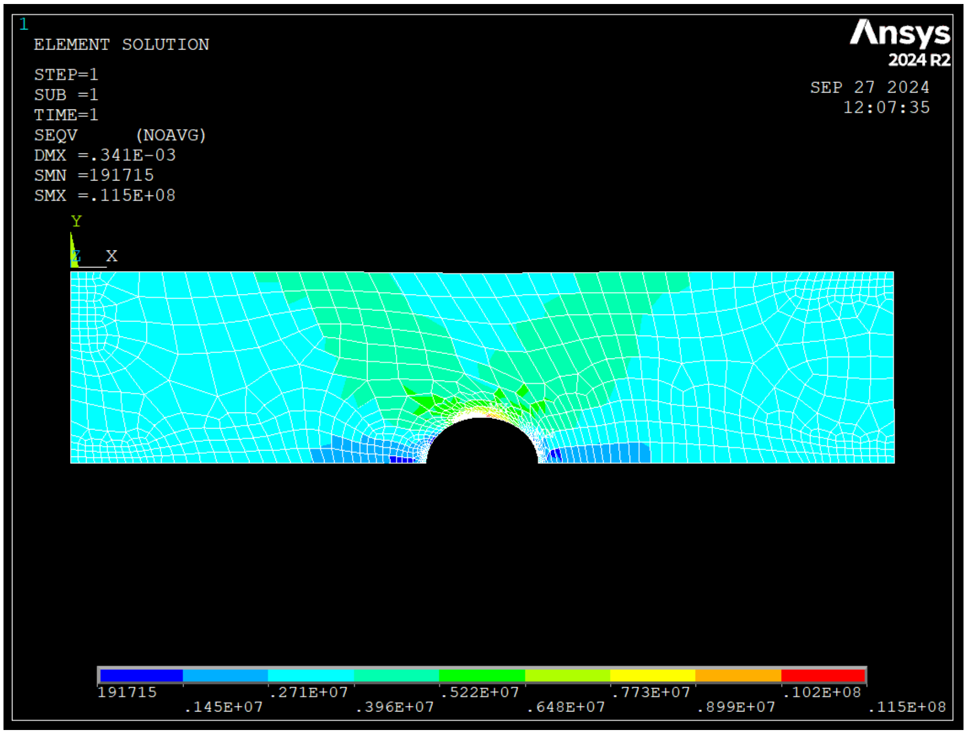


Figure 1.6. Von-Mises stress (Pa)

3. Analysis

Final stress concentration factor:

$\sigma_{nominal}$ = (applied load) / (cross-sectional area) = 12000 N / (0.01 m * 0.3 m) = **4.00e6 Pa**

σ_{max} = **1.29e7 Pa**

K = σ_{max} / $\sigma_{nominal}$ = 1.29e7 Pa / 4.00e6 Pa = **3.23**

Theoretical stress concentration factor:

K_t = 1 + 2 * (whole diameter) / (plate width) = 1 + 2 * (75 mm) / (600 mm) = **1.25**

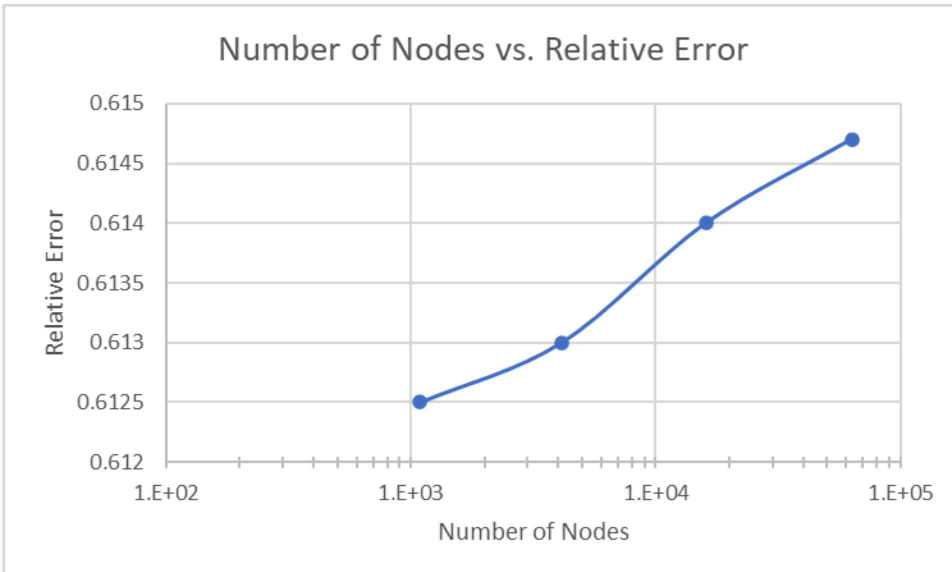
The final SCF is more than 2 times greater than the theoretical SCF.

4. Error Estimation

Theoretical σ_{max} = K_t * $\sigma_{nominal}$ = 1.25 * 4.00e6 Pa = **5.00e6 Pa**

Relative error = | ((theoretical value) - (simulation value)) / (simulation value) |

	Number of Nodes	Max stress (Pa)	Relative error
1st refinement	1090	1.2904e7	0.6125
2nd refinement	4123	1.2920e7	0.6130
3rd refinement	16021	1.2953e7	0.6140
4th refinement	63145	1.2978e7	0.6147

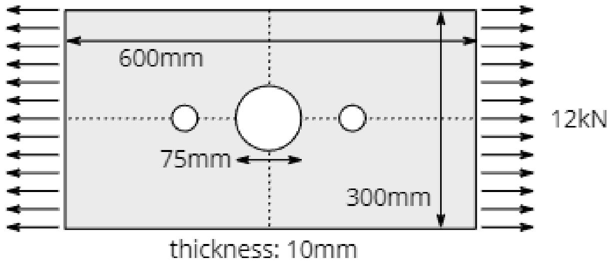


We can observe that the maximum stress does not vary a significant value after increasing the refinement (less than 1% difference between values). Thus, it is reasonable to say that the final reported solution is sufficiently accurate.

Part 2 Problem Statement

2 Stress Relief

Consider the following plate that is identical to the plate above, except with two additional holes added on either side of the central hole to relieve stress.



No dimensions are given here for the hole radius or spacing from the center hole: it is up to you to determine the configuration that optimally reduces stress.

Your analysis must contain the same elements as that for the first part.

5. Computational Model

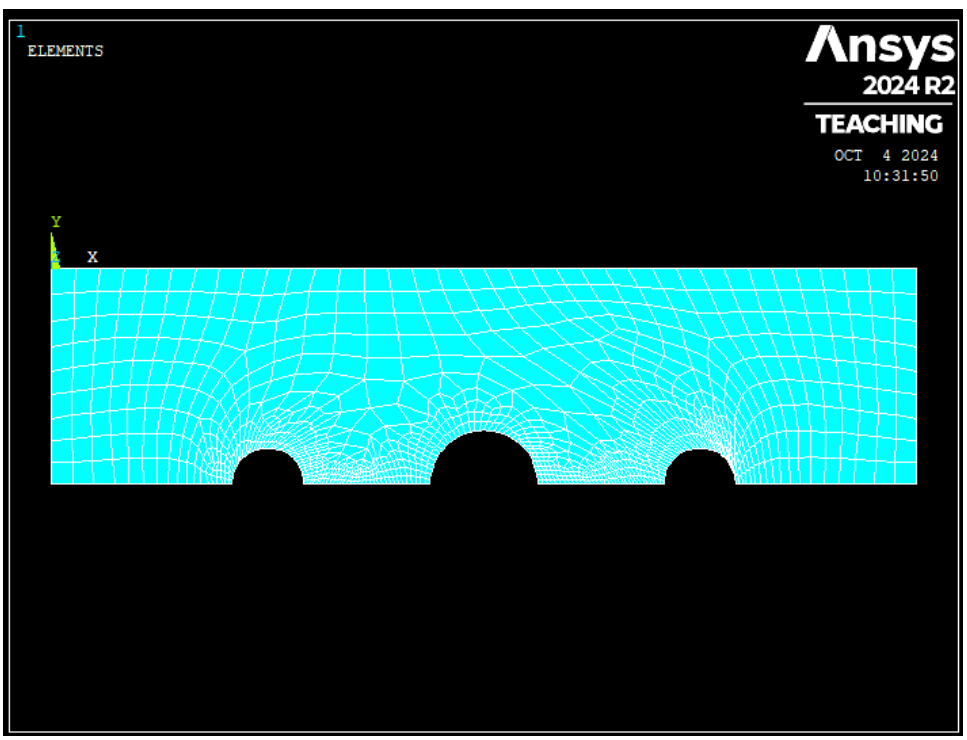


Figure 2.1. Plate modeled with mesh grid

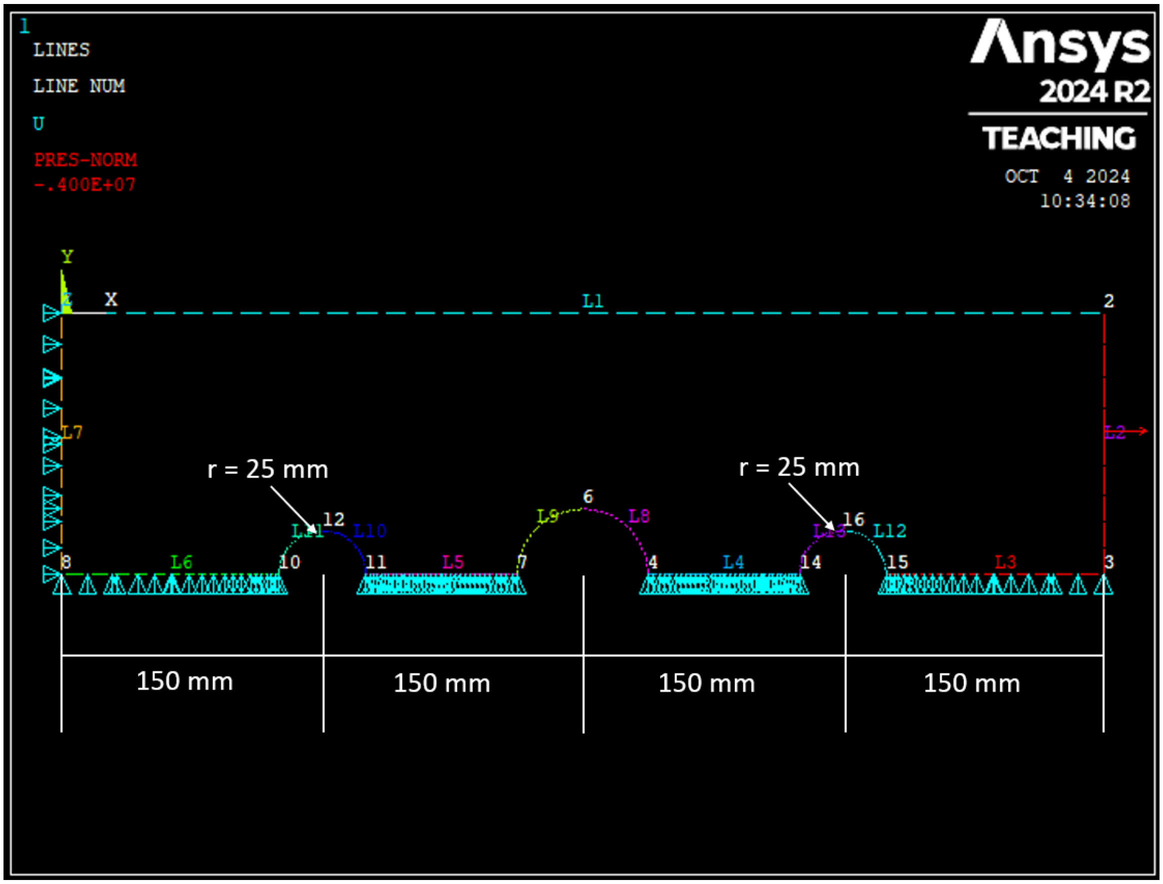


Figure 2.2. Plate constraints and applied loads

For our plate model with holes, we used the same element type from part 1, Plane182.

The chosen model was achieved by optimizing the model through a trial and error process. The radius of the added holes is 25 mm. They are symmetrically positioned, each having their centerpoint 150 mm from the outer edge. The optimization analysis details will be provided in the results section.

6. Results

The contour plots shown are the results from the design that had performed the best.

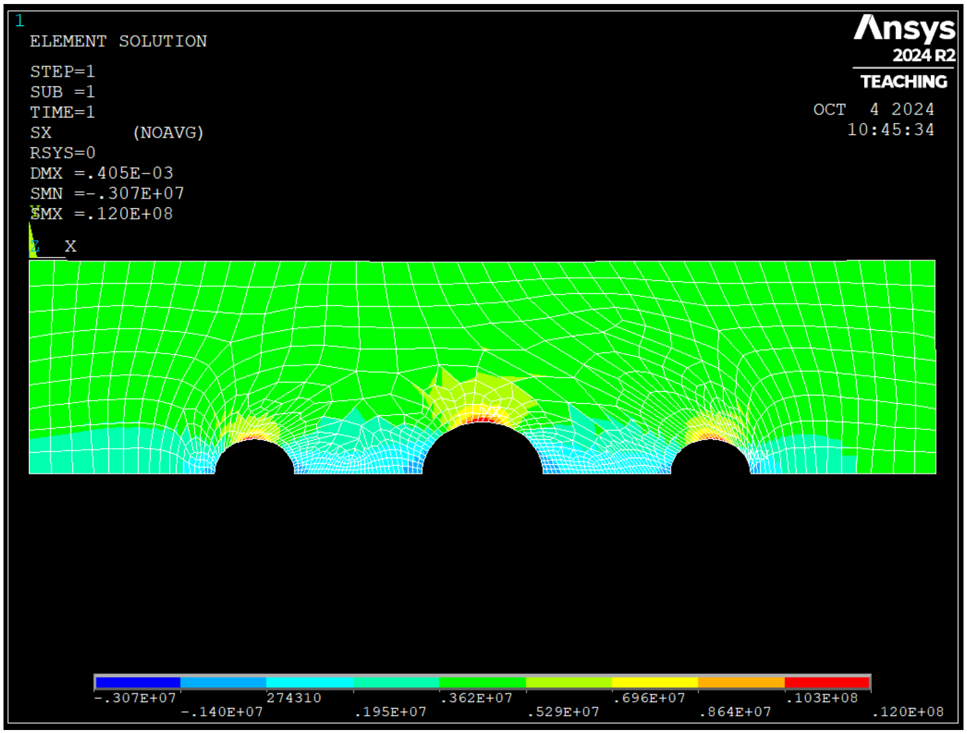


Figure 2.3. Axial stress in X-direction (Pa)

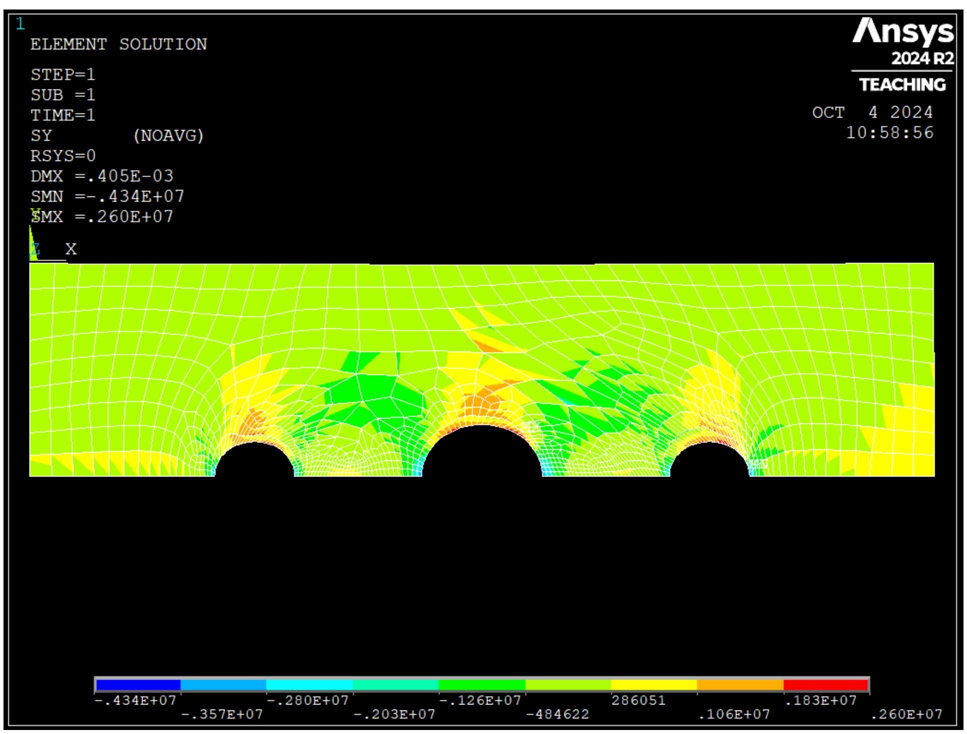


Figure 2.4. Axial stress in Y-direction (Pa)

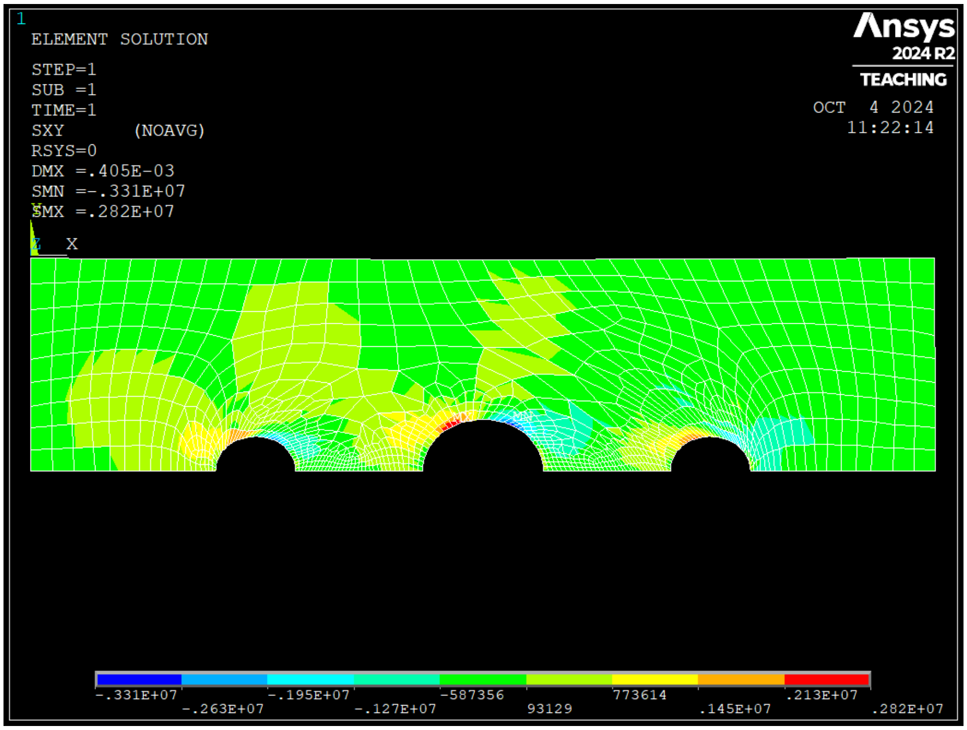


Figure 2.5. Shear stress (Pa)

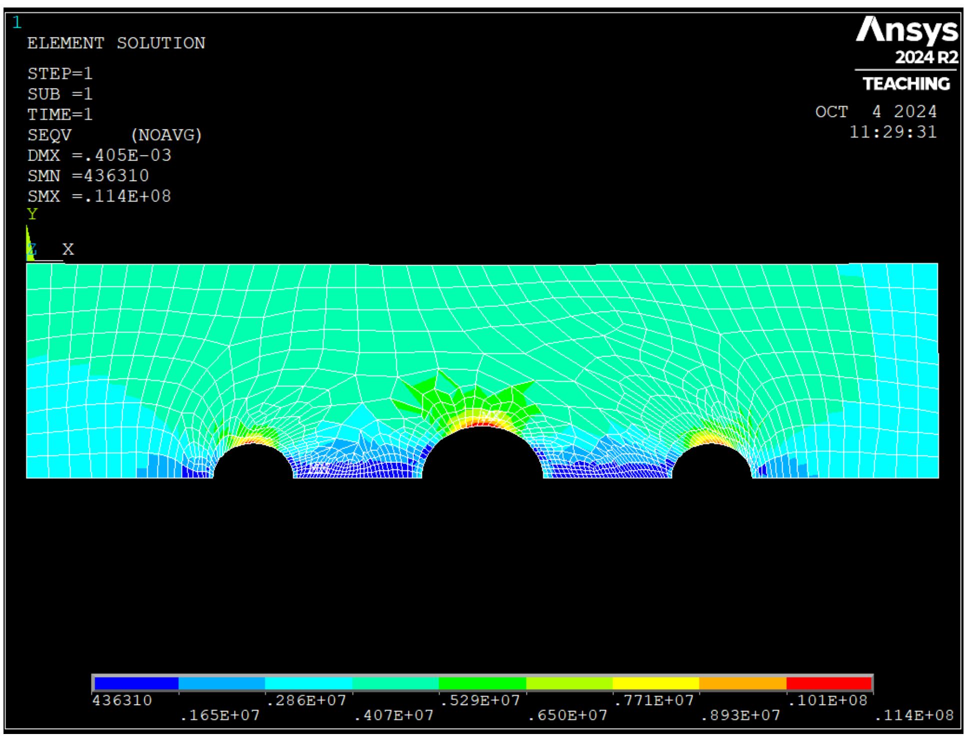


Figure 2.6. Von-Mises stress (Pa)

For optimizing the model we did this by completing a trial and error process. We chose to constrain the centerpoint of the added holes for this process and optimize the size of the added holes. We ran three tests, each with different sizes of the added holes. In test 1 the added holes had a radius of 25 mm, In test 2 the added holes had a radius of 12.5 mm, and In test 3 the added holes had a radius of 37.5 mm. Below is a table displaying the maximum values for the σ_{xx}, σ_{yy}, σ_{xy}, and the von mises stress. We also included the results for no added holes.

	σ_{xx} (Pa)	σ_{yy} (Pa)	σ_{xy} (Pa)	Von Mises (Pa)
No added holes	0.129e8	0.366e7	0.418e7	0.115e8
Test 1 (r=25mm)	0.120e8	0.260e7	0.282e7	0.114e8
Test 2 (r=12.5mm)	0.126e8	0.279e7	0.363e7	0.120e8
Test 3 (r=37.5mm)	0.132e8	0.212e7	0.266e7	0.125e8

7. Analysis

Final stress concentration factor:

$$\sigma_{nominal} = \mathbf{4.00e6\ Pa}$$
$$\sigma_{max} = \mathbf{1.20e7\ Pa}$$

$$K = \sigma_{max} / \sigma_{nominal} = 1.20e7\ Pa / 4.00e6\ Pa = \mathbf{3.00}$$

Theoretical stress concentration factor:

$$K_t = \mathbf{1.25}$$

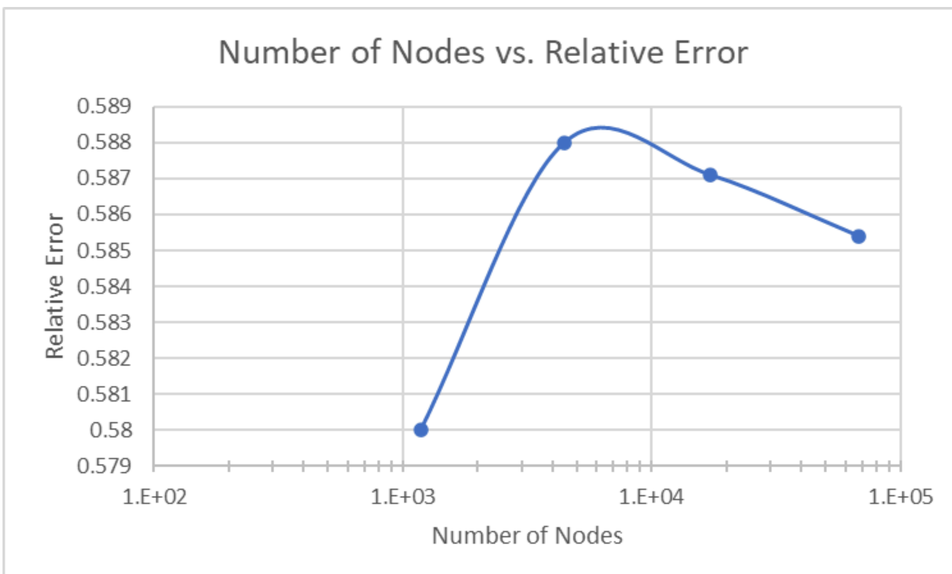
Like in part 1, the final SCF is more than 2 times greater than the theoretical SCF. However, their difference decreased, since the maximum stress decreased due to the new holes placement.

8. Error Estimation

Theoretical σ_{max} = **5.00e6 Pa**

Relative error = | ((theoretical value) - (simulation value)) / (simulation value) |

	Number of Nodes	Max stress (Pa)	Relative Error
1st refinement	1179	1.1905e7	0.5800
2nd refinement	4449	1.2135e7	0.5880
3rd refinement	17265	1.2110e7	0.5871
4th refinement	68001	1.2061e7	0.5854



Like in part 1, the maximum stress did not vary a significant value after increasing the refinement (less than 2% difference between values). Thus, the final reported solution is sufficiently accurate.